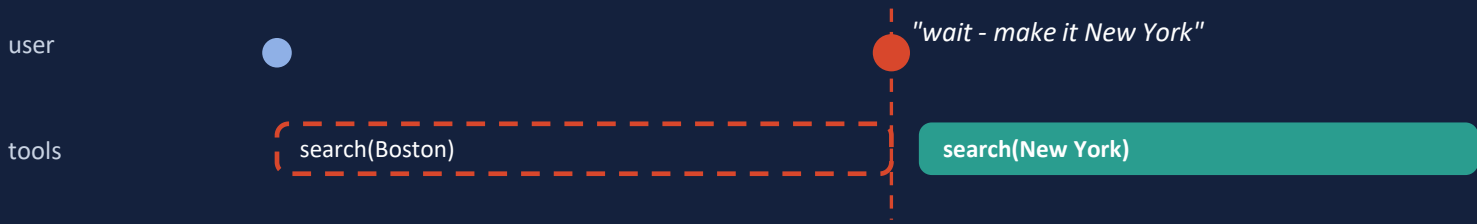


DUET

Making voice agents safe to interrupt

Samsung PRISM GenAI Hackathon 3.0 · Theme 05: Interruptible Real-Time Agents

Team SE7EN · SRM



Full-duplex speech is solved. Full-duplex action is not.

1

"Book a flight to Boston"

The agent starts searching - and may go on to book.

2

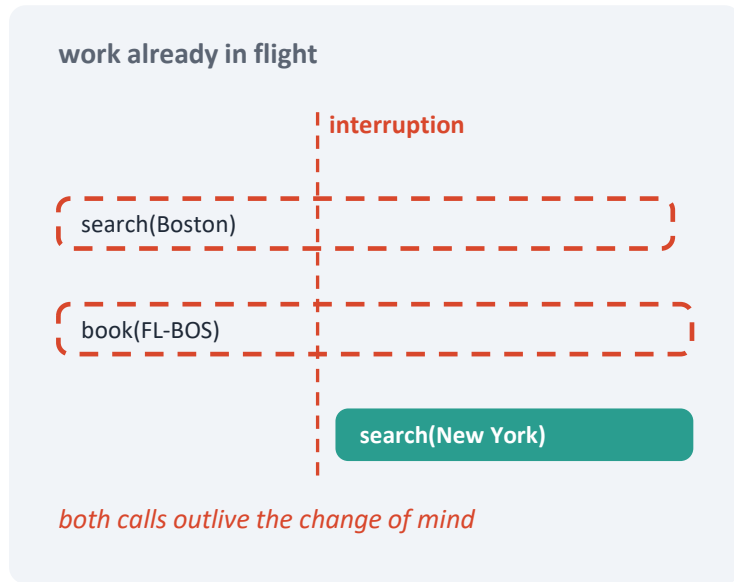
"Wait - make it New York"

The user changes their mind while that work is in flight.

3

Without coordination

The Boston result is still spoken; a booking can still commit - or commit twice.



Samsung's harness grades exactly this: stale work, state after the interruption, latency, and duplicate commits.

Four guarantees, held together

1

Never silent

A truthful, content-aware reply tens of milliseconds after every turn or interruption - however slow the real work.

2

Always interruptible

In-flight work is invalidated mid-execution; its results are never spoken and never committed.

3

Never double-commits

No irreversible action runs twice - not under retries, cancellations or re-plans.

4

Honest about uncertainty

When perception is unsure it asks, and names what: "Sorry, did you say Austin?"

No shipping assistant guarantees all four at once. The harness measures each of them.

Architecture: one dispatcher, one exit

DISPATCHER · runtime.py

synchronous · never awaits slow work · < 20 ms per event

FAST PATH

repairs, extraction,
acknowledgments

COORDINATOR

epochs (M1), ledger
+ gate (M2)

SLOW PATH

speech, frames, tools

STATE

slots + provenance
(M3), undo (M6)

EMITTER · the only queue write

snapshot · validation · no repeats · no premature claims

events in →

→ actions out

Two invariants, enforced by tests

The dispatcher never blocks.

Slowest handler 0.47 ms against a 20 ms budget.

Every action leaves through one function.

put_nowait appears exactly once in the engine.

Six mechanisms



Epoch-versioned state

An interruption starts a new epoch; everything conceived earlier is orphaned in one step.



Reversibility-gated commitment

Read-only work fires freely; irreversible work passes a gate and an idempotency ledger.



Slot provenance

Each value remembers where it came from, so a correction rewrites one slot and nothing else.



Perception-ahead

A frame is read when it arrives; a question about it waits for the reading, not the voice.



Calibrated abstention

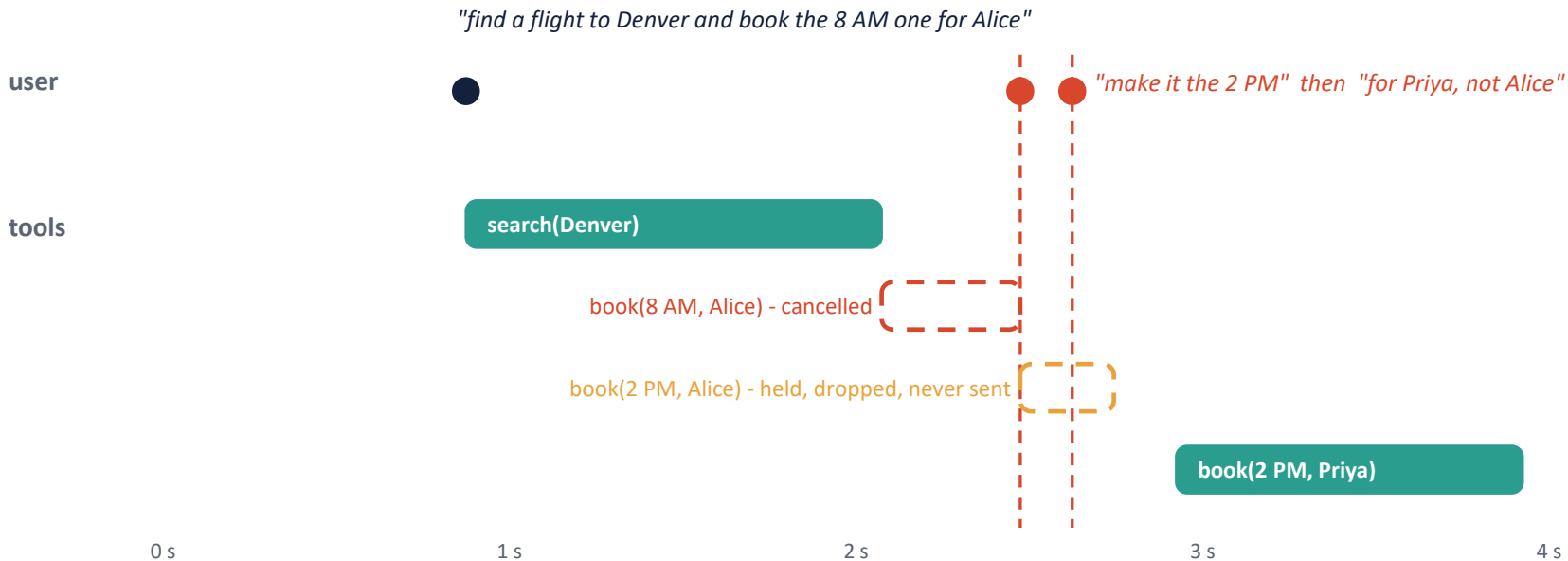
Confidence per word, not per sentence; a doubted value is confirmed by name.



Conversational undo

"Go back to what I said" restores an epoch checkpoint.

M1 in action: two corrections during a booking



Every computation carries the epoch it was conceived under. The second correction lands inside the 250 ms hold on irreversible work, so the held booking is dropped before it is ever sent. One booking goes out - 2 PM, for Priya - and "not Alice" never enters the state.

Tools it has never seen, handled honestly

A schema DUET met a moment ago

```
reserve_table (state_modifying)
  restaurant : string, required
  party_size : number, required
```

"Could you get us a table at The Olive Room for six, around 7 PM?"

```
reserve_table(restaurant="The Olive Room",
              party_size=6)
```

"All set - that went through, reservation reference RS-0042."

1

No tool names in the engine

Chosen and bound from the schema alone; chaining falls out of which arguments can be filled.

2

A timeout is not a failure

"...timed out, so I cannot tell whether it went through. I will not send it again without checking with you."

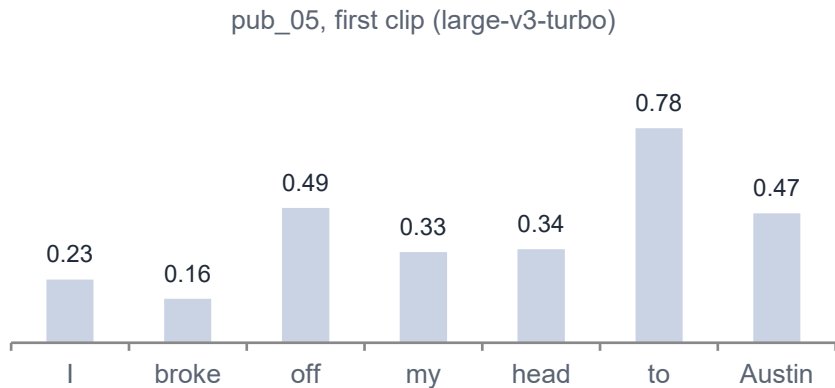
3

Promises, never premature claims

"Booking flight FL-DEN-8AM now" before the tool reports; "booked" only after.

Hearing and seeing: measured, not assumed

Speech: confidence per word



"Austin" at 0.47 is below the 0.65 bar, so DUET asks by name: "Sorry, did you say Austin?" - then searches Boston when the user says so.

Vision: a model we tested and rejected

full frame

"USB-C port"

centre crop

"HDMI port"

lower half

"USB-C port"

mirror

"USB-C port"

Qwen2.5-VL-3B named pub_07's HDMI port correctly on 1 of 4 variants and invented a printed label to match. A wrong answer fails two checks instead of one, so vision reading ships off: DUET says it cannot tell, names the candidate pages, and asks.

Results

97.4

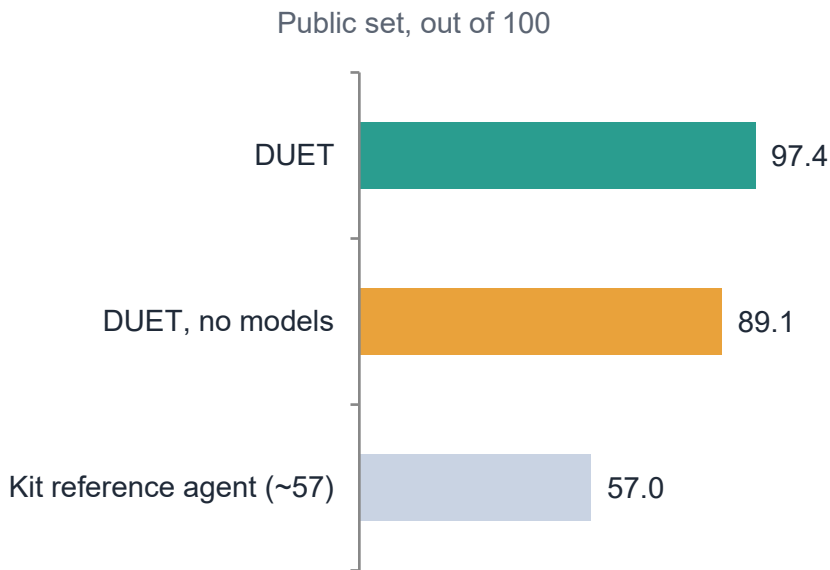
official weighted score,
public set (evaluator, scale 1)

99.7

mean on 120 scenarios, 8
templates, a seed never used
before

89.1

with every model switched off -
never silent, never crashes



0 crashes · 0 protocol errors · 0 abandoned calls across every run. 373 tests.

How we know: testing the tests

1

Conformance

23 scenarios the kit does not cover. Each interruption case is proven to fail an agent that does not cancel.

2

Transcript lint

The score cannot hear the agent. The first read found a garbled final answer in 6 of 17 scenarios that all scored 100. Now 0 of 32.

3

Our own chaos

Randomized paraphrase, pivot, booking and unseen-tool templates: 88.1 on first run, six general fixes, 99.7 on a seed never used before.

4

Degradation + pins

89.1 with no models. Every dependency and model pinned; the GPU stack proves itself in setup() or falls back.

Rule we kept: if a change improves the public nine but worsens unseen scenarios, it is a hardcode and it is reverted.

Honest limits, and where DUET goes next

Limits we measured

- Naming a component in a photo: the 3B vision model was wrong 3 times in 4, so it ships off.
- A self-repair whose trigger word is lost in heavy noise can leave the abandoned value in place.
- The planner is rules, not a language model: deterministic and fast, but its phrasing coverage is what we have tested.

Next, as a PRISM worklet

- DUET as the coordination layer between Bixby's speech and SmartThings actions - any device manifest, zero retraining.
- A larger vision model, admitted only once it passes our vision bench on device photos.
- A language-model planner raced behind the rules with a hard timeout - rules stay the floor.

Every limit is written up with its evidence in notes/FINDINGS.md (F18, F21), next to what we tried.

Full-duplex speech is solved.

DUET makes full-duplex action safe.

stale work: cancelled

irreversible: held, then once

the answer: grounded

Team SE7EN · SRM · entry point agent:ParticipantAgent · README.md reproduces every number